

EASTERN INTERNATIONAL UNIVERSITY

BECAMEX BUSINESS SCHOOL



COURSE: SCLM 479 INTEGRATED LOGISTICS

AND SUPPLY MANAGEMENT

**REPORT: STRATEGIC SUPPLY CHAIN & PURCHASING
ANALYSIS (2026–2032)**



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FINAL REPORT: STRATEGIC SUPPLY CHAIN & PURCHASING ANALYSIS (2026–2032)

Company: Rebisco Vietnam Ltd.

Focus: Navigating Volatility, Scaling Growth, and Sustaining Margins.

CHAPTER I: EXECUTIVE SUMMARY

Objective: Analyze operational strategies to position Rebisco as the manufacturing hub for the Indochina region by 2030.

Key Challenges:

- Fluctuating raw material prices (Cocoa, Palm Oil)
- Risks associated with single-source supply
- Pressure to increase production by 50% by 2032

Key Message: Transition from "Passive Procurement" to "Strategic Portfolio Management" (Kraljic-based) and optimize S&OP.



Figure 1: SWOT analysis

CHAPTER II

PART 1: DEMAND & SUPPLY PLANNING

I. Demand And Supply Planning for the 2028 “Matcha Doowee Donut” launch(Question 1)

1. Market Forecast and Matcha Price Volatility (2024 – 2030)

Planning for 2028 must consider growth indicators and global supply risks:

Growth Trends: The global Matcha market is expected to grow at a CAGR of 7.9% to 9.2% from 2024 to 2030 (Grand View Research, 2024; Mordor Intelligence, 2024).

2028 Price Forecast: Due to a shortage of Grade A supply from Japan and climate change impacts in key regions, Matcha prices are projected to rise by 12–15% in 2028 compared to 2025 levels (Expert Market Research, 2024).

Impact: High pressure on Cost of Goods Sold (COGS) requires a "Volume Reservation" strategy to lock in prices early through formal contracting with suppliers.

2. Launch Volume and Target Setting

2.1. Volume Estimation

Typically, a new flavor introduction (Line Extension) is expected to capture approximately 8% – 10% of the total category volume during its first year of launch.

Calculation: $6,637.58 \text{ MT} \times 10\% = 664 \text{ MT/year}$

2.2. Monthly Allocation (Seasonality)

The Vietnam confectionery market is highly seasonal. To ensure availability while managing inventory risks, the volume is allocated as follows:

- *Tet Season (Jan – Feb):* Accounts for approximately 30% of annual volume (~200 MT). This is the peak period for gifting and festive consumption.
- *Mid-Autumn Festival (Aug – Sept):* Accounts for approximately **15%** of annual volume (~100 MT).

- *Off-Peak Months:* The remaining volume will be distributed evenly at approximately 45 MT/month.

Table 1: Implementation Details & Key Target

Phase	Strategic Action	Implementation Details & Key Targets
1.Demand Forecasting	Proxy-based Modeling & Seasonal Adjustment	Volume Target: Based on the total 2028 Doowee category forecast of 6,637.58 MT, the Matcha variant is targeted at 10% market share, totaling ~664 MT/year. Seasonal Allocation: Concentrating 30% of annual volume in Q4 2027 & Q1 2028 to capture the peak Tet (Lunar New Year) demand.
2.Supply & Sourcing	Volume Reservation & Price Locking Strategy/	Cost Control: With Matcha prices projected to rise by 12-15% in 2028, Purchasing must secure a Blanket Purchase Agreement by late 2026 to lock in 2025-level pricing and guarantee Grade A supply from Japan. Strategic Buffer: Increase raw material safety stock to 4-6 months (vs. the standard 1 month) to insulate against climate-related supply disruptions.

3. Production & Operations	Agile Manufacturing & Freshness Focus	Small Batch Production: Utilize shorter production runs to ensure maximum freshness (shelf-life quality) and reduce the financial impact of potential expiry risks for a new flavor. Demand Shaping: Align with Marketing for "Pipe-fill" launches in Tier 1 & 2 mini-marts to test real-time Sell-out rates before full-scale expansion.
4. Risk Management	Expiry Control & VA/VE Integration	Early Warning System (EWS): Trigger immediate trade promotions (e.g., "Buy 2 Get 1") when inventory reaches 1/3 of its remaining shelf-life. Value Engineering: Collaborate with R&D to optimize secondary ingredients (fats/sugars) to offset the 15% price hike in Matcha without compromising the premium taste profile.

Implementation Timeline (2027 - 2028)

- **Q3 2027 (Preparation):** Finalize the 664 MT forecast. Execute "Volume Reservation" contracts with Japanese suppliers to hedge against the 15% price surge.
- **Q4 2027 (Pre-Launch):** Receive the first shipment of Matcha powder. Begin production for the Tet festive season "Pipe-fill" stock.
- **Jan 2028 (Market Launch):** Official product rollout. Monitor Weekly Sales Reports to adjust February and March production schedules dynamically.
- **Q2 2028 (Review):** Evaluate supply chain resilience. If scarcity persists, activate R&D's Value Analysis protocols to refine the cost structure.

Key Takeaway: The success of Matcha Doowee 2028 relies on Seasonal Forecasting, Early Price Hedging, and Agile Batch Production.

2.3. Estimated Holding Inventory Costs (4–6 Months Buffer)

Step 1: Key Inputs

Table 2: Estimated values for calculation

Parameter	Example Value
Annual Forecast Volume	664 MT
Monthly Usage	$664 \div 12 \approx 55$ MT/month
Buffer	4 months $\rightarrow$ 220 MT 6 months $\rightarrow$ 330 MT
Matcha COGS	\$10,000/MT (assumed)
Inventory Holding Cost Rate	12% per year

Step 2: Calculate Holding Costs

Formula:

$$\text{Inventory Holding Cost} = \text{Inventory Value} \times \text{Annual Holding Rate} \times \text{Number of Months} / 12$$

- **4-month buffer:**

$$220 \text{ MT} \times 10,000 = 2,200,000 \text{ USD}$$

$$\text{Holding Cost} = 2,200,000 \times 12\% \times (4/12) = 88,000 \text{ USD}$$

- **6-month buffer:**

$$330 \text{ MT} \times 10,000 = 3,300,000 \text{ USD}$$

$$\text{Holding Cost} = 3,300,000 \times 12\% \times (6/12) = 198,000\text{USD}$$

Step 3:

Table 3: Summary Table

Buffer (months)	Inventory Volume (MT)	Inventory Value (USD)	Holding Cost (USD)
4	220	2,200,000	88,000
6	330	3,300,000	198,000

Key Takeaway: Increasing buffer reduces risk of shortage but significantly increases holding costs. Trade-offs must be evaluated.

2.4. 2nd Sourcing / Alternative Plan for Matcha (Considering Taiwan)

Step 1: Selection Criteria

Quality: Grade A, flavor profile compatible with Doowee

Supply Capacity: MOQ meets 1–2 months production

Certifications: HACCP, ISO, Halal (if export needed)

Step 2: Supplier Options

Table 4: Selection Criteria for Suppliers

Source	Country	MOQ (MT)	Lead Time	Certification	Notes
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Primary	Japan	5 MT/lot	30 days	HACCP, ISO	High price, climate risk
Secondary	Taiwan	3 MT/lot	25 days	HACCP, ISO	Comparable quality, 5–10% lower cost

Step 3: Integration into S&OP / Procurement Strategy

1) Volume Reservation Split

Japan: 70% forecast volume (466 MT)

Taiwan: 30% forecast volume (198 MT) → mitigates risk

2) Contracting & Price Locking

Japan: Blanket Purchase Agreement → lock 2026-level price

Taiwan: Optional contract → flexible if Japan price spikes

3) Buffer / Safety Stock

Taiwan stock can provide 1–2 months safety stock to absorb supply disruptions

4) VA/VE Consideration

If Japan Matcha price rises >15%, increase Taiwan proportion in production without impacting flavor profile

Step 4: Combined Executive

Table 5: Combined Executive Table

Material	Total Forecast Volume 2028 (MT)	Primary Supplier	Secondary Supplier	Buffer Stock (MT)	Estimated Holding Cost (USD)
Matcha	664	Japan (70%)	Taiwan (30%)	4–6 months	88,000 – 198,000

Key Insight:

- Japan ensures premium flavor; Taiwan reduces supply risk and provides flexibility.
- Holding inventory must be balanced with cost; 4–6 months buffer covers peak season and potential shortages.

Table 6: Assumptions for Calculation

Parameter	Assumed Value
Projected Matcha demand 2028	664 MT/year
Japan vs. Taiwan allocation	60% Japan, 40% Taiwan
Matcha Grade A 2026 price (baseline)	Japan: 35 USD/kg, Taiwan: 28 USD/kg
Projected 2027 spot price	Japan: 39 USD/kg (+12%), Taiwan: 31 USD/kg (+10%)
Holding period	4–6 months (0.33 – 0.5 year)

Inventory holding cost	20%/year of inventory value
Early purchase for buffer	100% of 6-month demand

Step 1: Calculate Matcha quantity for 6-month holding

- 6-month demand = $664 \text{ MT} / 2 = \mathbf{332 \text{ MT}}$
- Japan/Taiwan allocation:
 $\text{Japan} = 332 \times 60\% \approx \mathbf{199 \text{ MT}}$
 $\text{Taiwan} = 332 \times 40\% \approx \mathbf{133 \text{ MT}}$

Step 2: Calculate inventory holding cost (6-month buffer)**Formula:**

Holding Cost = Inventory Value $\times$ Holding Rate $\times$ Holding Time

1. Japan:

- Inventory value = $199,000 \text{ kg} \times 35 \text{ USD/kg} = 6,965,000 \text{ USD}$
- 6-month holding cost = $6,965,000 \times 20\% \times 0.5 \approx \mathbf{696,500 \text{ USD}}$

2. Taiwan:

- Inventory value = $133,000 \text{ kg} \times 28 \text{ USD/kg} = 3,724,000 \text{ USD}$
- 6-month holding cost = $3,724,000 \times 20\% \times 0.5 \approx \mathbf{372,400 \text{ USD}}$

Total holding cost: $696,500 + 372,400 \approx \mathbf{1,068,900 \text{ USD}}$

Step 3: Calculate cost if purchasing in 2027 (spot price)

- Projected 2027 spot prices: Japan 39 USD/kg, Taiwan 31 USD/kg
- Total purchase cost = $199,000 \times 39 + 133,000 \times 31 \approx 7,761,000 + 4,123,000 = \mathbf{11,884,000 \text{ USD}}$

If purchased 6 months early + holding cost:

- Purchase at 2026 price:

Japan: $35 \times 199,000 = 6,965,000$ USD

Taiwan: $28 \times 133,000 = 3,724,000$ USD

- Total cost = $10,689,000 + \text{holding cost } 1,068,900 = \mathbf{11,757,900 \text{ USD}}$

Purchasing 6 months early and accounting for holding costs saves approximately **126,100 USD** compared to waiting to buy at projected spot prices in 2027, while also mitigating supply risk.

Table 7: Matcha Procurement Cost Comparison (Japan + Taiwan, 664 MT demand, 2028)

Procurement Option	Japan (199 MT)	Taiwan (133 MT)	Total (USD)	Notes
Hold 6 months (buy early 2026 price + holding cost)	$6,965,000 + 696,500 =$ 7,661,500	$3,724,000 + 372,400 =$ 4,096,400	11,757,900	Locks in price, mitigates supply shortage risk, allows 4–6 month buffer
Buy spot 2027	$199 \times 39 =$ 7,761,000	$133 \times 31 =$ 4,123,000	11,884,000	Higher risk of price surge, no buffer for shortages



Figure 2: Procurement cost comparison

II. Solving Bottlenecks and Seasonality (Q8 & Q9)

1. Question 8

In summer 2026 (May to August), the chocolate coating (enrobing) line is expected to become the main bottleneck in the factory. The enrobing process involves coating products with liquid chocolate and then passing them through a cooling tunnel, where the chocolate must solidify properly to ensure product quality. This cooling stage is critical because chocolate requires controlled temperature conditions to crystallize and form a stable coating (Food Coating Technologies, n.d.).

Under normal conditions, the cooling system can efficiently remove heat and maintain a stable production speed. However, during hot summer months, higher ambient temperatures reduce cooling efficiency and increase the time required for chocolate to solidify. Since cooling is a time-dependent process, longer solidification time forces the conveyor speed to decrease in order to maintain product quality (New Food Magazine, 2013). In addition, cooling tunnels are a key determinant of line efficiency and throughput, meaning any limitation in cooling capacity directly

reduces overall production output (BAF Engineering, n.d.). As a result, the enrobing stage becomes the slowest step in the production process, creating a bottleneck that limits the entire system's capacity. Consequently, output is expected to drop by approximately 25–30% compared to normal conditions.

According to the 2026 production forecast, total output of Doowee Donut is expected to exceed 6,000 tons, including more than 2,700 tons of chocolate flavor, over 2,000 tons of strawberry, and more than 1,200 tons of assorted products. In total, products requiring chocolate coating (including Extreme and Whoopie) account for nearly 11,000 tons, almost half of the factory's total production. Additionally, more than 3,400 tons of coating fat will be required. If demand peaks during the summer period, as is typically the case, the enrobing line will not have sufficient capacity to meet demand.

This issue is further intensified by conflicting objectives between departments. Sales aims to maximize product availability during peak snack seasons, while Production focuses on maintaining efficiency, minimizing waste, and controlling energy consumption. Increasing production speed under high-temperature conditions may lead to poor chocolate setting and quality issues, while reducing speed to protect product quality may result in stockouts and lost sales. Therefore, the company faces a clear conflict between demand and production capacity. If not properly managed, this bottleneck will hinder the company's ability to achieve its growth targets for 2026–2032.

Recommended solutions

a) Production Adjustment

The primary technical solution to address the enrobing line bottleneck during the hot summer months (May to August) is to shift all chocolate coating operations to the night shift (8 p.m. – 8 a.m.). High daytime temperatures in Vietnam increase the risk of chocolate melting during the coating process, which slows line speed, raises defect rates, and reduces overall production efficiency. By moving runs to cooler night conditions, the cooling tunnel can operate more effectively, allowing chocolate to set properly and consistently. This adjustment does not require new equipment but leverages environmental optimization, resulting in a significant increase in throughput, more stable operations, and fewer product defects. Additionally, scheduling chocolate

enrobing at night allows the daytime shifts to focus on uncoated products, such as crackers, creating a more balanced utilization of the plant's overall capacity.

b) Cross-functional Coordination

While technical adjustments improve line efficiency, solving the bottleneck fully requires closer coordination between departments. The Sales, Production, and Purchasing teams must engage in monthly Sales & Operations Planning (S&OP) meetings to align forecasts with production capabilities. During these meetings, Sales provides detailed peak-season demand plans, while Production evaluates the realistic throughput of the enrobing line, factoring in seasonal efficiency reductions caused by high temperatures. Purchasing ensures that critical raw materials, such as cocoa, chocolate, and fats, are available in sufficient quantity to prevent production stoppages. By establishing this structured planning process, Rebisco Vietnam can avoid overcommitting sales, prevent stockouts, and reduce the risk of product loss due to expiry or quality issues. This cross-functional approach also fosters proactive problem-solving and continuous improvement, strengthening internal communication and ensuring that production decisions are informed by both operational realities and market demand.

The combination of night shift production and cross-functional S&OP coordination enables the enrobing line to run closer to its maximum effective capacity without compromising quality. The factory can reliably meet seasonal peaks in demand, such as for Doowee Donut during summer months, while minimizing defective products. Furthermore, these solutions improve operational predictability, reduce pressure on operators during high-temperature periods, and enhance overall collaboration across departments, contributing to more strategic and data-driven decision-making.

2. Question 9: Aggregate Planning Strategy for Tet 2027

2.1. Context Evaluation

Demand is expected to increase sharply by approximately 300% during the Tet season, creating a short-term demand spike. The production system relies on tunnel ovens, which consume a large amount of energy and require continuous operation (24/7) to maintain efficiency. Frequent shutdowns or temperature adjustments, as required in a Chase strategy, may lead to energy waste and potential equipment damage. The main product, crackers in tin boxes, has a long shelf life

(more than one year). This characteristic makes it suitable for inventory accumulation before the peak season.

2. Data Assumptions and Model Explanation

To support the analysis, a simplified quantitative model is developed using Python. The purpose of this model is to compare the cost and behavior of different aggregate planning strategies under a demand shock scenario.

2.1. Demand Assumptions

The demand data is constructed based on the expected seasonal pattern:

- Base demand is assumed to be 1,000 MT per month under normal conditions.
- During the Tet season (January and February), demand increases by 300%, reaching 4,000 MT per month.

Therefore, the demand profile is defined as:

- October: 1,000 MT
- November: 1,000 MT
- December: 1,000 MT
- January: 4,000 MT
- February: 4,000 MT

This reflects a typical short-term demand spike observed in the FMCG industry during major holidays.

2.2. Cost and Capacity Assumptions

The following parameters are used in the model:

- Unit production cost: 1,000 USD per MT
- Inventory holding cost rate: 1% per month

- Overtime premium: 1.5 times the normal labor cost
- Normal production capacity: 1,000 MT per month

These assumptions are used to approximate real-world conditions:

- Holding cost represents storage, capital cost, and risk of obsolescence
- Overtime premium reflects higher labor cost during peak periods
- Capacity reflects the normal operating level of the production system

2.3. Strategy Modeling Logic

Level Strategy

- Production is kept constant across all months
- Production rate is calculated as the average demand over the planning horizon
- Inventory is accumulated in low-demand periods and used during peak demand

Cost considered:

- Inventory holding cost only

Chase Strategy

- Production is adjusted to match demand exactly each month
- No inventory is carried

Cost considered:

- Overtime cost when demand exceeds normal capacity

Hybrid Strategy

The Hybrid strategy is designed based on operational feasibility:

- October–November: Slightly above normal capacity (1,200 MT)
- December: Moderate increase (1,500 MT)

- January–February: Higher production (2,500 MT), supported by inventory and overtime

This structure reflects a gradual ramp-up approach, rather than sudden changes.

Cost considered:

- Inventory holding cost
- Overtime cost

2.4. Inventory Calculation Logic

Inventory is calculated dynamically using the following relationship:

$$\text{Inventory (t)} = \text{Inventory (t-1)} + \text{Production (t)} - \text{Demand (t)}$$

- Positive inventory represents stock available
- Negative inventory is avoided in the model (no backorder assumption)

2.5. Model Purpose

This model is not intended to provide exact financial results, but to:

- Illustrate the trade-offs between strategies
- Compare cost structures under different approaches
- Support decision-making with quantitative evidence

2.6. Summary of Python Model Results

The Python-based analysis provides a data-driven approach to evaluate and compare different aggregate planning strategies under a demand surge scenario.

Table 8: The summarized estimated total costs for each strategy

Strategy	Total Cost (Est)	Risk Level	Operational Feasibility
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Level	90,000	Medium	High (Stable oven operation)
Chase	3,000,000	High	Low (Operational stress on ovens)
Hybrid	1,965,000	Low	High

The results show clear differences between the three strategies:

The Level strategy has the lowest cost (90,000) because it avoids overtime. However, this comes at the expense of very high inventory levels, which may lead to storage pressure and reduced product freshness.

The Chase strategy has the highest cost (3,000,000) due to heavy reliance on overtime during peak months. In addition, it creates significant operational stress on the production system, especially the tunnel ovens.

The Hybrid strategy results in a moderate total cost (1,965,000). While it is more expensive than the Level strategy, it significantly reduces both inventory risk and overtime pressure, making it more practical for real operations.

3. Limitations of Pure Strategies

Level Strategy

A pure Level strategy requires the company to produce at a constant rate and build inventory well in advance.

However, this approach has several limitations:

- High inventory holding cost
- Increased warehouse space requirement
- Reduced product freshness after long storage periods

Therefore, although the Level strategy is cost-efficient in production, it creates significant inventory risks.

Chase Strategy

A pure Chase strategy adjusts production to match demand in each period.

This approach also presents several challenges:

- Requires extensive overtime (OT) or temporary labor
- Leads to potential worker fatigue and quality issues
- Significantly increases labor cost due to OT premiums
- Not suitable for tunnel ovens, which require stable operations

Therefore, while the Chase strategy is flexible, it is costly and operationally inefficient in this context.

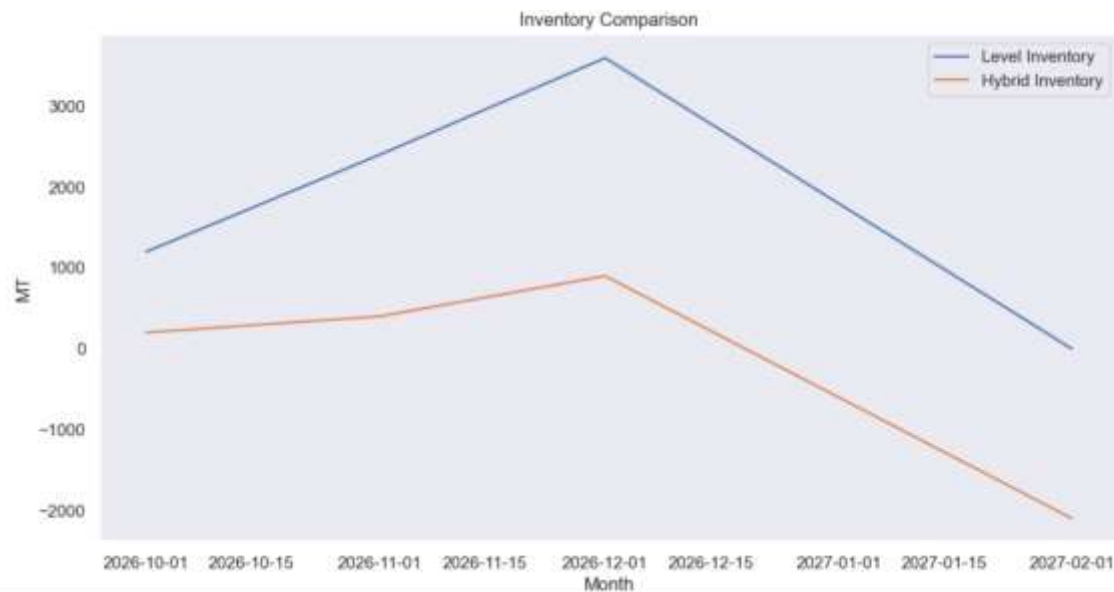


Figure 3: Inventory comparison

4. Proposed Solution: Hybrid Strategy & Economic Analysis

A Hybrid strategy combines the stability of the Level approach with the flexibility of the Chase approach.



Figure 4: Aggregate planning strategy comparison



Figure 5: Strategic production planning phases

Total cost includes:

- Inventory Holding Cost
- Labor Cost (regular and overtime)
- Energy Cost

The Hybrid strategy optimizes total cost because:

- It maintains stable production, reducing energy inefficiency
- It distributes overtime across multiple periods, reducing labor cost pressure
- It controls inventory levels, avoiding excessive holding cost

5. Recommendation & insights

It is recommended that the company adopts a Hybrid strategy, supported by subcontracting when necessary. If actual demand exceeds the forecast (e.g., more than 300%), the company can outsource simple processes such as packaging. This allows the main production system to remain focused on core operations.

Expected outcomes include:

- Maintaining a high service level (approximately 98%)
- Optimizing total operational cost
- Protecting equipment performance and workforce sustainability

The analysis highlights several key insights:

- There is no single optimal strategy. Each approach involves trade-offs between cost, flexibility, and operational feasibility.
- The main challenge is not only demand variability, but also production constraints, especially the requirement for stable operation of tunnel ovens.

- The Hybrid strategy is effective because it reduces extreme risks associated with both Level and Chase strategies.
- Inventory should be viewed as a strategic buffer, rather than a cost burden, as it helps absorb demand fluctuations.
- The objective of aggregate planning is not to minimize a single cost component, but to optimize total cost across the entire system.

PART 2: SOURCING & CATEGORY MANAGEMENT

I. Supply Risk Management(Q2 & Q3)

- *Breaking the Single-Source Trap (Q2):*

1. Problem assessment

Rebisco Vietnam is facing high supply risk because several important raw materials are approved from only one supplier or one exact product code, such as skim milk powder, whey powder, shortening, non-PHO fat, lactose powder, and cocoa powder. This weakens purchasing power, reduces negotiation leverage, and increases the risk of disruption, especially when demand rises or the approved supplier faces capacity, logistics, or price issues. The case also shows that the root problem is not only supplier availability, but also the interface between Purchasing and R&D: R&D specifications are too narrow, many materials are tailor-made, and R&D lacks resources to test alternatives.

2. Recommended solution

The best response is not just “find another supplier”, because the real bottleneck is the approval system. Rebisco should launch an Alternative Material Validation Program led jointly by Purchasing, R&D, QA, and Finance.

a. Break the R&D bottleneck

Purchasing should work with R&D to create an annual pipeline of high-risk raw materials for re-validation. Priority should be given to the most critical single-sourced items first, such as

cocoa powder, skim milk powder, and specialty fats. A cross-functional KPI can be introduced, for example:

- Number of alternative materials tested per year
- Number of second sources approved
- Percentage of spend under dual sourcing

This is consistent with the case's own raw material approval flow, where supplier identification, documentation, R&D testing, and QA audit all have to be completed before commercial approval.

b. Apply a 70/30 sourcing strategy

For materials where a second source can be qualified, Rebisco should use a 70/30 allocation model:

- 70% volume to the main supplier to secure scale discounts and stable quality
- 30% to the backup supplier to maintain competition and emergency flexibility

This reduces dependence without fully sacrificing price efficiency. It also creates a live backup source instead of a “paper-approved” supplier that is never used.

c. Standardize where possible

For some materials, the team should challenge whether the current specification is unnecessarily narrow. If the product does not truly require a highly customized input, Rebisco should shift from tailor-made materials to more standardized commodity-grade inputs. That would:, widen the supplier pool, reduce MOQ problems, improve cost competitiveness, shorten sourcing lead time.

d. Maintaining a Backup Plan and Relationships with Alternative Suppliers

During the long-term contract period with the main supplier, Rebisco needs to proactively develop 2-3 backup suppliers through framework agreements, ensuring they are ready to increase volume when needed without having to re-tender from scratch (GmbH, 2025).

Maintain regular contact with backup suppliers through quarterly meetings, sharing demand forecasts, and placing small trial orders to maintain a "hot" relationship – similar to the case of Ba Huân with eggs (large supplier but with Vifood as a replacement in case of market rumors) (Bnews, 2022).

Implement a policy of "transparent and trustworthy communication": Share production plans 3-6 months in advance, provide timely feedback on quality, and fulfill payment commitments to build long-term trust (Fmit, 2025).

Avoid situations of being forced to accept emergency pricing by establishing an "emergency clause" in the backup contract, allowing for the activation of an alternative source at a fixed price for the first 30 days, while also periodically evaluating supplier performance through KPIs such as delivery reliability (OTIF >95%) (Fmit, n.d.). This approach helps Rebisco shift from single-source risk to a flexible network, minimizing disruption without complicating the R&D approval process.

3. Strategic recommendation

In procurement terms, Rebisco should move from technical single-sourcing to controlled dual-sourcing. The goal is not to approve many suppliers for appearance, but to build a sourcing structure that protects continuity, improves bargaining power, and still preserves product quality.

4. Suggested conclusion sentence

Therefore, the best sourcing strategy is to treat single-sourcing as a cross-functional risk issue, not only a purchasing issue. Rebisco should institutionalize alternative-material validation, standardize specifications where feasible, and gradually shift key raw materials from single-source dependence to a dual-source model.

- **Egg Crisis (Q3):**

1. Problem assessment

The egg shortage is a business continuity crisis, not just a temporary price problem. In this case, Rebisco depends on fresh eggs for premium product quality and cannot easily switch to egg powder. At the same time, supply is hit by multiple shocks: cross-border trade disruption, post-

flood supply reduction, seasonal Tet demand, and the financial weakness of a major supplier. This creates a direct risk to production continuity, procurement budget, and COGS.

2. Recommended solution

Rebisco should activate a Business Continuity Plan (BCP) for eggs immediately.

a. Urgently diversify the supplier base

Depending on only Ba Huan and Nhien Poultry is too risky under current conditions (VnBusiness, n.d.). Purchasing should immediately scout and pre-qualify additional suppliers with acceptable scale and food safety capability, such as larger integrated livestock and food groups or satellite farm networks in nearby provinces. For example, C.P. Vietnam operates an integrated 3F+ model and lists eggs among its food products, while Dabaco is a major domestic agribusiness group with large-scale food and livestock operations. Ba Huan itself is a large national egg supplier, which shows why losing capacity from such a supplier can strongly affect the market (CPV Food, n.d.).

b. Move from distributor dependence to direct farm contracting

A strong medium-term solution is to sign direct long-term supply agreements with farms or farm clusters, instead of relying only on major commercial distributors. This is a partial form of vertical coordination. It can include:

- Guaranteed off-take contracts
- Minimum volume commitments
- Agreed pricing formula
- Quality and certification requirements
- Emergency allocation clauses during peak shortage periods

This will improve supply visibility and reduce dependence on financially stressed intermediaries.

c. Risk-Based Supply Chain Stratification

Divide eggs into three classes:

Core: Large, approved suppliers.

Contingency: Nearby backup farms in Binh Duong/Dong Nai.

Emergency: Short-term safety stock (because fresh eggs are perishable).

Because eggs are perishable, large inventories are not feasible – prioritize supplier diversification and contracts over stockpiles. Cost Analysis: Holding Cost vs. Downtime Cost

According to the Vietnam Logistics Report 2020, holding costs for inventory account for 11% of warehousing + 21% of loading and unloading + 8% of packaging, totaling 15-30% of the annual value of goods. For FMCG businesses like confectionery (with revenues of trillions of VND/year), holding fresh eggs for one week (assuming 10 billion VND) costs approximately 0.3-0.6 billion VND in holding costs (Meksmart, 2022).

Conversely, a one-day production stoppage results in significantly higher downtime costs: the global food industry loses \$4,000-\$30,000 per hour (~100-750 million VND per hour), and Vietnam faces similar losses due to supply chain disruptions (e.g., textiles/footwear experiencing raw material shortages, increased costs of 2,000 VND/kg, and lost deadline orders). For Rebisco (a large-scale manufacturer), one day of downtime is estimated to cost hundreds of millions to billions of VND (based on daily revenue of ~2-3 billion VND from industry reports) (CafeF, 2022).

d. Consider product priorities and demand

If there is a severe shortage, S&OP should temporarily prioritize fresh egg SKUs that generate the highest value/margin, reducing the volume of low-value products to strategically allocate eggs.

3. Strategic recommendation

Switch from purchasing at the usual cost to a resilience-focused sourcing model. In a supply crisis, continuity is more important than unit price (DPS Media, n.d.).

4. Suggested conclusion sentence:

In short, Rebisco views the egg shortage as an emergency continuity, responding through BCP, rapid supplier diversification, direct farm contracts, and S&OP allocation – better protecting

production than waiting for market prices to stabilize, because downtime costs outweigh holding costs (Johnson, 2026).

II. Question 4 & 5

I. Cacao (Q4 - Nhóm Strategic):

1. Classifying Cocoa in the Kraljic Matrix

Classification: Cocoa is classified as a Strategic item in Quadrant 4 of the Kraljic Matrix.



Figure 6: Classifying Cocoa in the Kraljic Matrix

High profit/value impact: Cacao is a core ingredient in flagship products such as Doowee Donut, KrimStix, and Rebisco Extreme, so it directly affects product quality, taste, consistency, and raw material cost. This means any increase in cacao price can significantly reduce profit margin.

High supply risk: Global cacao supply is facing a long-term structural deficit due to aging cocoa trees and crop diseases in major producing regions in West Africa. At the same time, Rebusco currently depends on a single supplier, Olam, which increases supplier dependency and reduces sourcing flexibility if disruption happens.

2. Propose Three Mitigation Strategies

2.1 Technical Strategy: Value Engineering (VA/VE)

a. Core Objective – Yield Optimization:

Improve the chocolate enrobing process to minimize material giveaway and production waste as much as possible. The goal is to ensure that every gram of cocoa, which is currently very expensive, is converted into revenue in the most efficient way.

Specific implementation actions:

Precise control of operating parameters: Closely monitor and standardize critical process variables such as temperature, coating viscosity, and line speed. This helps prevent over-coating, which increases material consumption, or under-coating, which may lead to product defects.

Preventive maintenance: Regularly maintain and calibrate dosing systems and spray nozzles to ensure that equipment always operates at the highest possible level of accuracy.

In-process quality control (IPQC): Shift from passive finished-product inspection to continuous quality monitoring directly on the production line. This allows problems to be detected early, corrected quickly, and prevented from causing large-scale material waste.

b. Formula restructuring – Use of Cocoa Butter Substitute (CBS):

Implementation: Work directly with the R&D team to increase product formulation flexibility. In particular, partially replace pure cocoa butter with CBS in chocolate coating applications, such as for products like Doowee.

Effectiveness: This solution can significantly reduce material costs in a market where cocoa is facing a structural deficit and prices remain above USD 6,000 per ton. The key point is

that CBS can still maintain strong functional performance, ensuring that the coating keeps its gloss, sets properly, and does not reduce the chocolate profile that consumers expect.

2.2 Financial Strategy: Risk Hedging and Integrated Planning

a. Hedging and Flexible Forward Contracting

Execution: Use financial risk management tools to lock in prices through forward contracts when the market shows temporary dips. However, to reduce the risk of contract failure under extreme price volatility, these contracts should include a **Flexible Price Band Clause**. Under this arrangement, both parties agree on a safety margin, for example $\pm 1\%$ to 2% . If the actual market price moves within this range, a renegotiation mechanism will be triggered so that the price can be adjusted according to a pre-agreed risk-sharing formula.

Business impact:

Budget certainty: This approach helps protect profit margins during 2026–2027 despite the continued supply deficit in the cocoa market.

Contract survival and fairness: The 1–2% flexible band acts as a pressure-release mechanism within the contract. It creates a fairer and more sustainable agreement for both sides: Rebeco is protected from overpaying when the market falls, while the supplier is less likely to face severe losses or break the contract when cocoa prices rise sharply.

b. Open-Book Costing

Execution: Shift to an open-book costing model with strategic suppliers, such as Olam.

Business impact: This model allows the company to separate and understand the true material cost from other supporting costs such as logistics charges and premium mark-ups. As a result, it provides a stronger basis for transparent negotiations, where price adjustments are linked to actual changes in global market indexes rather than subjective supplier-driven price increases.

3.3 Supply Chain Finance Strategy: Leveraging Corporate Credibility to Secure Supply

The mechanism: Instead of directly using its own working capital to provide advance payments or financial support to suppliers, Rebeco can leverage its stronger credit profile as a

financing tool. The company would sign a three-year off-take agreement with clearly committed purchase volumes.

Execution: Cocoa suppliers, especially cooperatives or farms that have production capacity but limited access to capital, can use this long-term contract or purchase order from a large and credible buyer like Rebisco as a form of collateral. They can then bring the contract to a bank to obtain financing. Because the bank can see a more predictable future cash flow backed by Rebisco's purchasing commitment, it is more willing to provide loans at more favorable interest rates.

The supplier can then use this financing to cover key operating needs, such as buying fertilizer, paying workers, or investing in proper cocoa drying equipment to meet quality standards, without being forced by cash pressure to sell early at unfavorable prices.

Win-win impact:

For the supplier: This strategy helps solve the seasonal cash flow problem at the beginning of the crop cycle. As a result, suppliers can focus more on improving cocoa quality and output.

For Rebisco: The main advantage is that the company can preserve its own working capital, with no cash tied up in supplier pre-financing. Rebisco avoids the opportunity cost of advancing funds, while still locking in supply of a strategic raw material for the full three-year period. At the same time, this arrangement creates high switching costs, because the supplier becomes closely tied to Rebisco through a contract that also supports its financial survival.

III. Palm Oil (Q5 - Leverage/Strategic):

1. Practical Impact on Rebisco Vietnam



(Thị Trường Hàng Hóa, 2024)

Figure 7: Palm Oil Market Price Fluctuation

The practical impact on Rebisco Vietnam is not necessarily an immediate physical shortage of palm oil, because the B50 policy has not yet been fully implemented. However, the company may still face higher purchasing prices and sourcing costs, as Indonesia continues to prioritize domestic biodiesel demand and has already increased the export levy on palm oil. If Rebisco Vietnam depends too heavily on Indonesia, it will be exposed to country concentration risk, meaning that any policy change in Indonesia could directly affect raw material prices, delivery schedules, and budgeting stability. In addition, if crude oil prices continue to rise, Indonesia may decide to reinstate B50 in mid-2026. Therefore, the main risk for Rebisco Vietnam is not only physical supply disruption, but more importantly policy uncertainty and market volatility.

2. Practical Actions of the Purchasing Director

Measure 1: Reduce Dependence on Indonesia through Geographic Diversification

One practical action is to reduce Rebisco's dependence on Indonesia by diversifying palm oil sourcing. Buying 100% from Indonesia creates very high supply risk, especially under the B50

policy. Therefore, Rebisco should set a KPI to lower the sourcing share from Indonesia and build a more balanced supply structure, such as 70% Indonesia and 30% Malaysia. This helps the company reduce dependency on one country and maintain production continuity if Indonesian exports become more restricted.

Assumptions:

- Indonesia price = USD 938.87/MT (Rri, 2026)
- Malaysia price = MYR 4,496/MT (*MALAYSIA PRICES OF CRUDE PALM OIL*, n.d.)
- Exchange rate = 1 USD = 3.9531 MYR
- Malaysia price converted = $4,496 / 3.9531 = \text{USD } 1,137.34/\text{MT}$
- Import 1000 Tons/month

Table 9: Comparison of Total Cost by Import Ratio

Import Ratio	Indonesia (tons)	Malaysia (tons)	Total Cost (USD/month)
30:70	300	700	946,885.47
70:30	700	300	942,305.20
50:50	500	500	944,595.34
100:0	1000	0	938,870.00
0:100	0	1000	950,320.68

The 70:30 sourcing mix is preferred because it offers the best trade-off between cost efficiency and supply security. Although 100:0 has the lowest monthly cost, the savings are small compared with 70:30. By allocating 30% to Malaysia, the company reduces overdependence on Indonesia and lowers the risk of disruption from policy changes, export restrictions, or delivery instability. Therefore, 70:30 is a more balanced and practical sourcing option.

Measure 2: Pre-approve Backup Suppliers in Advance

Another practical action is to qualify backup suppliers before any disruption happens. Rebisco should not wait until there is a shortage to start searching for new suppliers. Instead, the Purchasing team should work early with QA and R&D to collect supplier documents such as specifications and COA, then conduct sensory testing and small pilot runs. Once the supplier meets the required standards, it should be pre-approved and recorded in the system, so the company can place orders immediately when needed.

Measure 3: Increase Short-Term Safety Stock

Another practical action is to build short-term safety stock for sensitive and difficult-to-replace materials. This creates a buffer to protect the company when the market faces sudden disruption. Rebisco should take advantage of temporary price declines to buy and store additional volume. However, because fats and oils can become rancid over time and affect product quality, the company should only keep enough stock for around 3 to 6 months. This level is more practical because it supports production continuity without locking up too much working capital.

Purchase Cost for Stockpiling:

$$\text{Purchase Cost} = \text{Number of tons} \times \text{USD } 938.87$$

Table 10: Purchase Cost of Palm Oil under Different Stockpiling Options

Stockpiling Option	Quantity to Be Stockpiled (tons)	Purchase Cost (USD)
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Stockpile for 3 months	3,000	2,816,610
Stockpile for 6 months	6,000	5,633,220

The table shows that purchase cost increases directly with the stockpiling volume. Stockpiling for 6 months requires twice the quantity and almost exactly twice the purchase cost of the 3-month option. This means the longer stockpiling plan provides stronger supply security, but it also requires significantly higher upfront cash and puts greater pressure on working capital.

Holding cost:

$$\text{Holding Cost per Month} = \text{Inventory Value} \times 2\%$$

The 2% monthly holding cost is an estimated average derived from the typical annual carrying cost range of 15%–30% of inventory value. Using 2% per month implies an annual rate of about 24%, which falls within this common range (*What Is Carrying Cost?* | Zoho Inventory, 2024).

Table 11: Monthly and Total Holding Cost of Palm Oil Inventory

Option	Inventory Value (USD)	Holding Cost per Month (2%)	Total Holding Cost
Stockpile for 3 months	2,816,610	56,332.20	168,996.60
Stockpile for 6 months	5,633,220	112,664.40	675,986.40

The table shows that once holding cost is included, longer stockpiling becomes much more expensive. Although the 6-month option provides stronger supply security, it also creates significantly higher inventory carrying cost and puts greater pressure on working capital. Therefore, the company must balance supply stability against the financial burden of holding excess inventory.

Rebisco Vietnam should choose the 3-month stockpiling option. This option provides a reasonable safety buffer against short-term supply and price volatility, while avoiding the heavy working capital burden and high holding cost of a 6-month stockpile. Since Indonesia has scrapped the B50 plan for 2026 and will remain at B40, the immediate risk is more about policy uncertainty and export levy-driven price increases than severe physical shortage. Therefore, a 3-month stockpile is the more balanced and financially prudent option.

IV. Total cost of ownership (question 6)

Wheat flour is a leverage item and is very important in many key products of Rebisco Vietnam, such as Doowee Donut, Rebisco Crackers, Whoopie, and Rebisco Extreme. In this analysis, wheat flour demand is estimated by focusing on BOM items in which it is explicitly specified as the primary ingredient, specifically in the following mixtures:

- *Wheat Flour, Sugar, Glucose, Water*
- *Wheat Flour, Yeast, Salt, Water*
- *Whole Wheat Flour, Honey*

These mixtures are selected because wheat flour is clearly the dominant component in the mixture, while other ingredients (e.g., sugar, yeast, water) act as secondary or functional additives.

Table 12: Wheat Flour Demand in 2027

BOM Ingredient	Demand 2027 (kg)
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Wheat Flour, Sugar, Glucose, Water	1,918,064
Wheat Flour, Yeast, Salt, Water	1,624,446
Whole Wheat Flour, Honey	295,791
Subtotal (Direct Wheat Flour)	3,838,301 kg

Wheat flour is assumed to account for approximately 70% of the total weight of these mixtures. This range reflects typical industrial formulations for bakery products, where flour serves as the primary structural component while other ingredients such as water, sugar, and fat contribute to texture and flavor.

Total Wheat Flour Demand Formula:

Wheat Flour Demand = total Direct Wheat Flour X wheat flour ratio assumption = 3,838,301 kg x 0.7 = **2,686,811 kg = ~2,687 ton**

The total demand is 2,687 tons in 2027, which represents the wheat flour category. This total is used for supplier selection. The main problem is to choose the supplier that gives the lowest total cost (TCO) while still ensuring reliable supply. Two suppliers are compared below:

Table 13: Comparison of 2 suppliers

Criteria	Supplier A (Local)	Supplier B (Imported)	Rating
Price	~8% higher	Lower	B

Lead time	3 days	45 days	A
Logistics cost	Very low	Higher (freight + tariff)	A
Inventory level	Low	High	A
Supply risk	Low	High (delay, global risk)	A
Flexibility	High	Low	A
Strategy fit	High (local sourcing)	Low	A

Formula:

Purchase Cost = Demand x Unit Price

- Purchase Cost (Supplier A) = 2,687 T x 286 .2 USD/ton = 769,019.4 USD
- Purchase Cost (Supplier B) = 2,687 T x 265 USD/ton = 712,055 USD

Logistics Cost=Demand×Logistics Cost per ton

- Landed cost is assumed at 20 USD/ton to include freight and import tariff. This is reasonable because logistics costs are usually around 5-10% of product value according to Organisation for Economic Co-operation (OECD, 2021).
- Logistics cost (Supplier A) = 2,687 x 0 (assumption) = 0
- Logistics cost (Supplier B) = 2,687 x 20 USD = 53,740 USD

Average Inventory=(Demand/2) x (Leadtime/365)

Holding Cost=Average Inventory x Unit Price x Holding Rate

- Holding cost (Supplier A) = 351.15 USD
- Holding cost (Supplier B) = 5688.64 USD

Risk Cost=Purchase Cost x Risk Rate

- A 3% risk premium is applied to reflect supply chain uncertainty and disruption risks associated with long lead times, consistent with global risk insights (World Economic Forum, 2023).
- Risk Cost (Supplier A) = 769,019.4 USD x 0 (assumption) = 0
- Risk Cost (Supplier B) = 712,055 USD x 3% = 21361.65 USD

TCO = Purchase + Logistics + Holding + Risk

Table 14: TCO results

Cost Component	Supplier A	Supplier B	Difference (A-B)
Purchase cost	769,019.4 USD	712,055 USD	+56,964.4 USD
Landed cost	0	53,740 USD	-53,740 USD
Holding cost	351.15 USD	5688.64 USD	5,337.49 USD
Risk premium	0	21,361.65 USD	-21,361.65 USD

Total TCO	769,370.55 USD	792,845.29USD	-23,474.74USD
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Even though Supplier B has a lower price, its total cost is 792,845.29 USD higher per year due to higher logistics, inventory, and risk costs.

Scenario 100% supplier A: Total TCO = 769,370.55 USD

Scenario 50% Local (A) / 50% Imported (B): Total TCO = (769,370.55 USD x 50%) + (792,845.29 USD x 50%) = 781,107.92 USD

Scenario 70% Local (A) / 30% Imported (B): Total TCO = (769,370.55 USD x 70%) + (792,845.29 USD x 30%) = 776,412.972 USD

Scenario 80% Local (A) / 20% Imported (B): Total TCO = (769,370.55 USD x 80%) + (792,845.29 USD x 20%) = 774,065.5 USD

Scenario 90% Local (A) / 10% Imported (B): Total TCO = (769,370.55 USD x 90%) + (792,845.29 USD x 10%) = 771,718.024 USD

Table 15: TCO scenario comparison

Scenario	Total TCO (USD)	Difference vs 100% A
100% Local (A)	769,370.55	Lowest
90% A / 10% B	771,718.02	+2,347
80% A / 20% B	774,065.50	+4,695
70% A / 30% B	776,412.97	+7,042

50% A / 50% B	781,107.92	+11,737
100% Imported (B)	792,845.29	+23,475

RECOMMENDED SOLUTIONS

Recommendation 1: Focus on Cost Savings

If the company wants to achieve the lowest total cost, the best option is to choose 100% Supplier A (Local Miller). This option gives the lowest TCO at USD 769,370.55, which is lower than Supplier B by about USD 23,475 and also lower than all mixed options. The main reason is that Supplier A has very low logistics cost, short lead time (3 days), and almost no risk cost. This helps reduce inventory and overall cost. It also matches Rebisco's strategy of using local suppliers for leverage items like wheat flour.

In addition, this option supports the company's goal to reduce COGS by 5% in 2027 and improves flexibility in planning and operations (S&OP). Therefore, Rebisco should consider signing a 6–12 month contract with Supplier A to keep the price stable.

Recommendation 2: Focus on Safety and Backup Supply

If the company wants to reduce risk and have a backup supplier, a better option is to use a 90% Supplier A / 10% Supplier B strategy. In this case, the total TCO is USD 771,718.02, which is only about USD 2,347 higher than using 100% Supplier A (around 0.3% difference). This small extra cost provides better supply security.

Supplier B can act as a backup source in case of problems such as supply disruption, natural disasters, or production issues at the local supplier. At the same time, using 90% Supplier A still keeps most benefits, such as low inventory, short lead time, and strong alignment with local sourcing strategy. This approach also supports the company's plan to build a strategic buffer for imported materials.

PART 3: FINANCE & R&D INTERFACE

I. Cost of Good Sold (Question 7)

The Finance team requires an exact 5% reduction in total raw material costs for 2027. This is a critical target because raw materials account for about 55-60% of our total production cost.

The challenge comes from price movement in two different groups:

- **Strategic materials** (Cocoa and some specialty fats) are expected to continue increasing due to global supply shortages.
- **Negotiable materials** (Sugar and Flour) are expected to stay stable or slightly decrease. These materials also have a large share in most product formulas.

If we do nothing and wait for Cocoa prices to rise, our costs will exceed the plan. However, if we take advantage of Sugar and Flour, which are easier to negotiate and have high usage, we can generate enough savings to offset the increase in Cocoa. In short, this is a clear opportunity: Use controllable materials to compensate for uncontrollable ones and achieve the 5% COGS reduction target.

Proposed solutions

We propose using bulk purchasing for Sugar and Flour and applying long-term contracts for Cocoa based on historical average prices.

1. Volume commitment with scheduled deliveries for Sugar and Flour

Based on material requirements and the 2027 production forecast, total demand for sugar and flour is estimated to exceed 6,300 tons per year, mainly driven by Doowee, Crackers, Extreme, and Whoopie product lines. Since these materials have a long shelf life and are relatively easy to store, Rebisco can consolidate the total demand and leverage higher purchase volumes to negotiate more favorable pricing.

To capture this advantage, the company will enter into a full-year contract, committing to the entire annual volume. This approach is expected to secure a 6–8% price discount compared to current market levels.

However, to avoid excessive inventory holding costs (12% annually), deliveries will not be made in a single bulk shipment. Instead, the total volume will be divided into smaller batches with bi-weekly (every 15 days) delivery schedules, ensuring a balance between cost efficiency and inventory control.

→ Expected outcome: Achieve 6–8% cost savings on sugar and flour, which can help offset the increasing costs of cocoa.

2. 3-year Cocoa contract based on historical average price

We calculated Cocoa prices based on historical data (2024–2026):

- Highest price: around 12,000 USD/ton
- Lowest price: around 2,900 USD/ton
- Average price: around 7,450 USD/ton



Figure 8: Historical cocoa price data (Source: Trading Economics – US Cocoa Futures Historical Data, updated March 26, 2026)

We will sign a 3-year contract with suppliers using the average price of 7,450 USD/ton. The contract will include a review clause: If prices increase sharply, both parties will review every 6–12 months. → This helps stabilize Cocoa cost, reduce risk, and improve cost forecasting.

By purchasing Sugar and Flour in bulk (taking advantage of long shelf life and easy storage) and signing a 3-year Cocoa contract at the historical average price of 7,450 USD/ton with adjustment terms, we can generate enough savings to fully offset Cocoa cost increases. As a result, we can achieve the 5% raw material cost reduction target for 2027 without affecting product quality or profitability.

II. VA/VE Value Analysis (Question 10)

1. Problem Assessment

R&D proposed using Natural Strawberry Extract (4x cost) to replace artificial flavor in Whoopie 2028, aiming for a "clean label" trend. Purchasing objected due to a sudden increase in COGS (flavor typically accounts for 1-3% of the cost of ingredients in baked goods, based on the case of Vietnamese cookies).

Risk: Customers may not recognize the difference (scent research shows that >70% of consumers cannot distinguish between natural and synthetic strawberry in blind tests due to >300 olfactory receptors but limited perception). Therefore, value is destroyed if the cost increases without a corresponding improvement in function (Nguyễn, 2022).

2. Value Analysis Process (5 Steps)

Step1: Function Breakdown

Table 16: Strawberry Flavor Function Analysis

Function	Artificial	Natural Extract	Natural-Identical

Taste/Aroma	Strawberry profile	Superior freshness	Equivalent
Clean Label Claim	"Artificial flavor"	"Natural strawberry extract"	"Natural & artificial flavors"
Shelf Stability	High	Medium	High

Step 2: Blind Sensory Test (100 consumers)

A: 100% artificial

B: 100% natural

Typical result: 65-75% indistinguishable (based on olfactory science research). → Equivalent function.

Step 3: Willingness-to-Pay (Post-Reveal)

After blind testing, reveal the label → Measure WTP premium:

- If <30% of customers recognize the product & WTP +5-10% of the price: Reject full conversion.
- If >50% prefer natural: Approve blend strategy.

Step 4: Quantitative Cost-Value Calculation

Realistic assumption (based on the case of Vietnamese biscuits: flavor ~0.3kg/1000 pieces, scaled to a batch of 100kg):

Artificial: 100 VND/kg → 10,000 VND

Natural: 400 VND/kg → 40,000 VND

Table 17: Quantitative Cost-Value Calculation

Blend Ratio	Total Cost (VND)	Cost Increase	Value Decision Logic
100% Artificial	10,000	0% (baseline)	Max value if blind test equivalent
30% Natural + 70% Artificial	22,000	120%	Optimal: "Natural blend" claim, manageable cost (+2% total Whoopie COGS)
50% Natural + 50% Artificial	25,000	150%	Trial premium variant (+5-7% price)
70% Natural + 30% Artificial	31,000	210%	Only if WTP >10% & sensory superior
100% Natural	40,000	300%	Reject unless function improves 3x (unrealistic)

Step 5: Decision Matrix

Table 18: Decision Matrix by Test Results

Test Result	Decision	Rationale
Customers cannot differentiate (70%)	30% Natural	Function = artificial, marketing gain "contains natural", cost +120% acceptable

Customers prefer natural (30%)	50% Natural	Balance function-cost, launch "Whoopie Natural" line (+7% price)
No WTP premium	Stay Artificial	Pure cost discipline

3. Strategic recommendation

Select 30% Natural + 70% Artificial as default:

Optimal Value: Function equivalent (blind test), reasonable clean label claim
 Cost Control: +120% flavor cost = +1.5-2% total COGS (flavor <2% raw materials)
 Risk Mitigation: 6-month trial, measure 5% sales uplift → scale full portfolio
 R&D-Purchasing Alignment: Compromise vs. deadlock

R&D validates blend formulation Q1/2028, purchasing sources bulk natural MOQ.
 Therefore, S&OP trial launch Tet 2028. Outcome: Successful Value Engineering – maintained function, optimized cost, improved market positioning.

PART 4: CONCLUSION & RECOMMENDATIONS

1. Conclusion

This report provides a comprehensive strategic analysis of Rebisco Vietnam's supply chain, procurement, and market operations for the 2026–2032 period. The findings indicate that the company is strategically well-positioned at its VSIP II-A facility to serve as a regional manufacturing hub. Achieving the target of increasing production by 50% by 2032 will require a shift from a "Passive Procurement" approach to Strategic Portfolio Management, ensuring a more proactive and resilient supply chain.

The analysis highlights key risks related to volatile raw materials such as Cocoa, Palm Oil, and Sugar. Single-source dependencies expose the company to potential supply disruptions and price fluctuations, which could constrain growth if not mitigated through supplier diversification and strategic sourcing frameworks such as the Kraljic Matrix. At the same time, emerging market

opportunities, particularly the launch of high-value products like the “Matcha Doowee Donut” (projected at 664 tons per year by 2028), present significant potential to increase market share. Capitalizing on this opportunity will require effective management of the anticipated 12–15% price surge in Matcha powder and careful coordination across production, logistics, and procurement.

Beyond supply chain considerations, the report emphasizes the importance of market trends, digitalization, and sustainability. Expanding into health-conscious and organic product lines aligns with rising consumer preferences, while leveraging omnichannel platforms such as TikTok Shop and Shopee can provide direct insights into consumer demand and bypass traditional distribution constraints. Logistics and digital transformation initiatives, including real-time warehouse management and tracking systems, are critical to supporting the scale-up of operations and positioning Rebisco Vietnam as an Indochina distribution hub.

In summary, Rebisco Vietnam has a strong foundation and favorable market positioning. Sustaining growth and operational excellence in the 2026–2032 period will depend on strategic sourcing, risk mitigation, demand planning, digital transformation, and market-responsive innovation. By addressing these areas, the company can transform current challenges into opportunities and secure a resilient, sustainable growth path.

2. Recommendations

To ensure sustainable growth and operational excellence, several key recommendations are proposed for Rebisco Vietnam:

2.1. Strategic Sourcing and Procurement

The company should implement the Kraljic Matrix to categorize all raw materials into Strategic, Bottleneck, Leverage, and Non-critical items. For strategic items such as Cocoa, long-term partnership contracts should be pursued to stabilize pricing. In addition, supplier diversification is critical to reduce dependence on single sources and protect against geopolitical or climate-related supply disruptions. Integrating sustainability criteria into supplier audits will further align procurement practices with global “Green Action” trends and future environmental regulations.

2.2. Demand Planning and S&OP Optimization

Enhancing the Sales and Operations Planning (S&OP) process is essential to manage seasonal production surges effectively, such as the 30% increase during Lunar New Year and the 15% increase for the Mid-Autumn Festival. Leaner inventory protocols for non-seasonal items should be adopted to reduce carrying costs and free up working capital, while maintaining strategic safety stocks for highly volatile raw materials.

2.3. Risk Management and Digital Transformation

For high-demand launches, such as the 2028 Matcha product, implementing volume reservation contracts will secure early access to critical supply quantities, mitigating the impact of forecasted global shortages. Concurrently, investments in logistics digitalization, including real-time tracking and warehouse management systems (WMS), will support the scale-up of the VSIP II-A plant as it transitions into an Indochina distribution hub.

2.4. Market Expansion

Rebisco Vietnam should leverage omnichannel integration by utilizing platforms such as TikTok Shop and Shopee to bypass traditional distribution bottlenecks and gather direct consumer insights for product development. Additionally, expanding the product portfolio to include health-conscious options, such as low-sugar or organic snacks, will capture the growing consumer preference for sustainable and healthy food products.

By implementing these recommendations, Rebisco Vietnam can strengthen its supply chain resilience, optimize operational efficiency, and capitalize on emerging market opportunities, thereby achieving sustainable growth in the 2026–2032 period.

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